4.1.9:

Solution:

Proof. Let $A, B \in \mathbb{M}_{2\times 2}(\mathbb{F})$ with

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

so

$$AB = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}.$$

Then,

$$\begin{aligned} \det(AB) &= (ae+bg)(cf+dh) - (af+bh)(ce+dg) \\ &= acef + adeh + bcfg + bdgh - (acef + adfg + bceh + bdgh) \\ &= adeh - adfg + bcfg - bceh \\ &= ad(eh-fg) - bc(eh-fg) \\ &= (ad-bc)(eh-fg) \\ &= \det(A)\cdot\det(B) \end{aligned}$$

which was to be shown. □

4.1.11*a*:

Solution:

Proof. Consider some $E \in \mathbb{M}_{2\times 2}(\mathbb{F})$ an elementary matrix given by applying at most one elementary row operation to I, the identity matrix.
Case 1 (no operation): If no operations are applied then clearly $E = I$ so

$$\delta(E) = \delta(I) = 1 = \det(I) = \det(E).$$

Case 2 ($r_1 \leftrightarrow r_2$): If the only operation applied is switching rows then

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = E$$

so $\det(E) = -1$. Now, using the facts that δ is a linear function of each row and that $\delta(A) = 0$ if any two rows of A are identical, we may consider

$$\begin{aligned} 0 = \delta\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right) &= \delta\left(\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\right) \\ &= \delta\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}\right) \\ &= \delta(I) + 0 + \delta(E) + 0 = 0 \\ &\Rightarrow \delta(E) = -1 = \det(E) \end{aligned}$$

Case 3 ($r_i \to kr_i$): In the case that a row is scaled, without loss of generality (since the case in which $i = 2$ is identical), we take $i = 1$ so $I \to E$ by $r_1 \to kr_1$ for $k \in \mathbb{F}$:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{r_1 \to kr_1} \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = E$$

so $\det(E) = k$. Then,

$$\delta\left(\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}\right) = \delta\left(\begin{pmatrix} k \cdot 1 & k \cdot 0 \\ 0 & 1 \end{pmatrix}\right) = k\,\delta\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = k\,\delta(I) = k = \det(E).$$

Case 4 ($r_i \to r_i + kr_j$): As before, without loss of generality we may take $i = 1$ and $j = 2$, as the argument is identical should they be swapped, and $k \in \mathbb{F}$ so $I \to E$ by $r_1 \to r_1 + kr_2$:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{r_1 \to r_1 + kr_2} \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} = E$$

so $\det(E) = 1$. But

$$\delta\left(\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}\right) = \delta\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & k \\ 0 & 1 \end{pmatrix}\right) = \delta(I) + k\delta\left(\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}\right) = 1 + 0 = 1 = \det(E).$$

Hence, $\det(E) = \delta(E)$ for all elementary 2×2 matrices. □

4.1.11∗:

Solution:

Proof. Let $A \in \mathbb{M}_{2\times 2}(\mathbb{F})$ so

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

So,

$$\begin{aligned}
\delta\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) &= \delta\left(\begin{pmatrix} a & 0 \\ c & d \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & b \\ c & d \end{pmatrix}\right) \\
&= \delta\left(\begin{pmatrix} a & 0 \\ c & 0 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix}\right) + \delta\left(\begin{pmatrix} 0 & b \\ 0 & d \end{pmatrix}\right) \\
&= 0 + ad + (-bc) + 0 \\
&= ad - bc = \det(A)
\end{aligned}$$

with the third step following from the second using identical techniques to the previous cases.
Thus, $\delta(A) = \det(A)$ for all $A \in \mathbb{M}_{2\times 2}(\mathbb{F})$. □

4.2.7:

Solution:

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix} = -6 + 3(-2) = \boxed{-12}$$

4.2.8:

Solution:

$$\begin{vmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \\ -1 & 3 & 0 \end{vmatrix} = -1 \begin{vmatrix} 0 & 2 \\ 1 & 5 \end{vmatrix} - 3 \begin{vmatrix} 1 & 2 \\ 0 & 5 \end{vmatrix} = -1(-2) - 3(5) = \boxed{-13}$$

4.2.14:

Solution:

$$\begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 0 \\ 7 & 0 & 0 \end{vmatrix} = 7 \begin{vmatrix} 3 & 4 \\ 6 & 0 \end{vmatrix} = 7(-24) = \boxed{-168}$$

4.2.18:

Solution: Consider the sequence of elementary row operations

$$A = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 2 & -5 \\ 3 & -1 & 2 \end{pmatrix} \xrightarrow{r_1 \to r_1 + r_2} \begin{pmatrix} 0 & 0 & -2 \\ -1 & 2 & -5 \\ 3 & -1 & 2 \end{pmatrix} = \tilde{A}$$

so $\det(A) = \det(\tilde{A})$. Then,

$$\det(A) = \det(\tilde{A}) = \begin{vmatrix} 0 & 0 & -2 \\ -1 & 2 & -5 \\ 3 & -1 & 2 \end{vmatrix} = -2 \begin{vmatrix} -1 & 2 \\ 3 & -1 \end{vmatrix} = -2(-5) = \boxed{10}$$

4.2.23:

Solution:

Proof. We will prove this via induction on n for $A \in \mathbb{M}_{n\times n}(\mathbb{F})$. Consider the first non-trivial value of n, $n = 2$ and let $A \in \mathbb{M}_{n\times n}(\mathbb{F})$ be any square upper-triangular matrix:

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

so $\det(A) = ac$ is the product of its diagonal entries. Thus, for $n = 2$ our proposition is true and we can now consider $n = 2$ as the base case. Now, suppose that for $n = k \geq 2$ the determinant of a

$k \times k$ upper-triangular matrix is the product of its diagonal entries. We must now show this is true for $n = k + 1$. Let $A \in \mathbb{M}_{(k+1)\times(k+1)}(\mathbb{F})$ and $\tilde{A} \in \mathbb{M}_{k\times k}(\mathbb{F})$ so that

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1(k+1)} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_{(k+1)(k+1)} \end{pmatrix} \quad \text{and} \quad \tilde{A} = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_{kk} \end{pmatrix}$$

where $A, \tilde{A}$ are both upper triangular. Now, we consider the determinant of A found by cofactor expansion across the last row (the cofactor sign is $(-1)^{2(k+1)} = 1$)

$$\det(A) = a_{(k+1)(k+1)} \begin{vmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_{kk} \end{vmatrix} = a_{(k+1)(k+1)} \det(\tilde{A}) = a_{(k+1)(k+1)} a_{kk} \cdots a_{22} a_{11}$$

by the induction hypothesis. Thus, the determinant of any upper triangular square matrix is the product of its diagonal entries. □

4.3.12:

Solution:

Proof. Let $Q \in \mathbb{M}_{n\times n}(\mathbb{F})$ be orthogonal such that $QQ^T = I$. Then,

$$\begin{aligned} \det(I) = \det(QQ^T) &= \det(Q) \cdot \det(Q^T) && \text{(Thm 4.7)} \\ &= \det(Q) \cdot \det(Q) && \text{(Thm 4.8)} \\ &= [\det(Q)]^2 = 1 \\ &\Rightarrow (\det(Q) - 1)(\det(Q) + 1) = 0 \end{aligned}$$

and hence either $\det(Q) = 1$ or $\det(Q) = -1$. □

4.3.15:

Solution:

Proof. Let $A, B \in \mathbb{M}_{n\times n}(\mathbb{F})$ such that A is similar to B. That is, there exists some invertible $P \in \mathbb{M}_{n\times n}(\mathbb{F})$ such that $B = P^{-1}AP$. But

$$\begin{aligned} \det(B) = \det(P^{-1}AP) &= \det(P^{-1}) \cdot \det(AP) && \text{(Thm 4.7)} \\ &= \det(P^{-1}) \cdot \det(A) \cdot \det(P) && \text{(Thm 4.7)} \\ &= \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) && \text{(Corr. to Thm 4.8)} \\ &= \det(A) \end{aligned}$$

thus $\det(B) = \det(A)$ as was to be shown. □

4.3.24:

Solution:

Proof. Let A be the matrix with -1 on the subdiagonal, last column $(a_0, \ldots, a_{n-1})^T$, and zeros elsewhere, and consider $\tilde{A} = A + tI$ so

$$\tilde{A} = A + tI = \begin{pmatrix} t & 0 & 0 & \cdots & 0 & a_0 \\ -1 & t & 0 & \cdots & 0 & a_1 \\ 0 & -1 & t & \cdots & 0 & a_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & a_{n-1} + t \end{pmatrix}.$$

Now, we will prove that

$$\det(\tilde{A}) = t^n + t^{n-1}a_{n-1} + \cdots + t^2 a_2 + t a_1 + a_0$$

via induction on n for all matrices $\tilde{A} \in \mathbb{M}_{n\times n}(\mathbb{F})$ such that $\tilde{A}$ is of the necessary form. Consider $n = 2$ as our first non-trivial value of n and hence our base case. Then,

$$\tilde{A} = \begin{pmatrix} 0 & a_0 \\ -1 & a_1 \end{pmatrix} + \begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix} = \begin{pmatrix} t & a_0 \\ -1 & a_1 + t \end{pmatrix} \Rightarrow \det(\tilde{A}) = t^2 + t a_1 + a_0$$

and so clearly the determinant of $\tilde{A}$ is in the desired form. Now, assume that for any $\overline{A} \in \mathbb{M}_{k\times k}(\mathbb{F})$ of the necessary form, $\det(\overline{A}) = t^k + t^{k-1}a_{k-1} + \cdots + t a_1 + a_0$. We must now show that for $A \in \mathbb{M}_{(k+1)\times(k+1)}(\mathbb{F})$, $\det(A) = t^{k+1} + t^k a_k + \cdots + t a_1 + a_0$. However,

$$\begin{aligned} \det(A) &= \begin{vmatrix} t & 0 & 0 & \cdots & 0 & a_0 \\ -1 & t & 0 & \cdots & 0 & a_1 \\ 0 & -1 & t & \cdots & 0 & a_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & a_k + t \end{vmatrix} \\ &= t \begin{vmatrix} t & 0 & 0 & \cdots & 0 & a_1 \\ -1 & t & 0 & \cdots & 0 & a_2 \\ 0 & -1 & t & \cdots & 0 & a_3 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & a_k + t \end{vmatrix} + (-1)^{(k+1)+1} a_0 \begin{vmatrix} -1 & t & 0 & \cdots & 0 \\ 0 & -1 & t & \cdots & 0 \\ 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & -1 \end{vmatrix} \\ &= t(t^k + t^{k-1}a_k + \cdots + t a_2 + a_1) + (-1)^k(-1)^{k+2} a_0 \qquad \text{(IH with } a_1, \ldots, a_k\text{; Problem 4.2.23)} \\ &= t^{k+1} + t^k a_k + \cdots + t^2 a_2 + t a_1 + a_0 \end{aligned}$$

which was to be shown. □

4.4.2*c*:

Solution:

$$\begin{vmatrix} 2+i & -1+3i \\ 1-2i & 3-i \end{vmatrix} = (2+i)(3-i) - (-1+3i)(1-2i) = (7+i) - (5+5i) = \boxed{2-4i}$$

4.4.3*c*:

Solution:

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix} = -6 + 3(-2) = \boxed{-12}$$

4.4.3*e*:

Solution:

$$\begin{vmatrix} 0 & 1+i & 2 \\ -2i & 0 & 1-i \\ 3 & 4i & 0 \end{vmatrix} = 3 \begin{vmatrix} 1+i & 2 \\ 0 & 1-i \end{vmatrix} - 4i \begin{vmatrix} 0 & 2 \\ -2i & 1-i \end{vmatrix} = 3(2) - 4i(4i) = 6 + 16 = \boxed{22}$$